Radio Mini Projects:

# Design of an antenna for decametric observations

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Timeline: 2nd May, 2025 - 24th June, 2025

## Aim

The project aimed at constructing an antenna for decametric (DAM) observations of outburst emissions from Jupiter/ Sun in the 18-28 MHz frequency range that occur mainly due to interactions between the planet's magnetic field and charged particles(which are linked with its moon Io). The idea was to design an antenna for the same which could capture these outbursts with sufficient accuracy.

## Objectives

The objective was to design and possibly build an antenna that could capture the DAM signals from Jupiter. The entire process would involve:

- Understanding the true nature of these outbursts, what causes them and why is it necessary to observe them in the first place.
- Selecting a specific kind of antenna that provides enough gain and directivity to be able to observe the outbursts and pinpoint where exactly they are coming from.
- Designing the antenna selected for the same using EM CAD software Ansys HFSS and determining if it would indeed work the way we want it to.
- Final production of the same if time permits and doing IRL observations of the same.

# Methodology

## Theoretical working of the entire model

The antenna was to be pointed at a spot on Jupiter and the 28 - 38 MHz band was to be observed by running multiple instances of the software SDRConsoleV3 or GNURadio to analyze multiple frequency points across the DAM band and observe fluctuations in their magnitude for an extended period of time. This can be set up via a Raspberry Pi as well to enable remote observations and check for fluctuations in the range. When the said fluctuations appear, those time points can be indications of some abnormal environmental activity on the surface of Jupiter such as a new storm which interacts with the magnetic field of the planet.

Taking into consideration the presence of multiple simultaneously occurring storms across the surface or due to small meteorites falling into the planet, hence creating a similar fluctuation. There are ways to determine the nature of the event such as if it is a meteorite, the disturbance would be for a short period of time such as 5-10 minutes for Earth and possibly lesser for Jupiter due to its thicker atmosphere. If the disturbance were to be caused by storms on the other hand, the overall effect of the same could last for a significantly longer period. Another thing to note would be that not all storms on the gassy giant would be powerful enough to cause such an effect on the magnetic field of the planet. Only storms with a significant magnitude would be able to show us the phenomenon and it could take several days if not months to observe the same (Luck is an important factor here).

Sometimes, the disturbances are caused due to Jupiter's moons Io, Europa and others proximity to the planet as well. That can be worked around by making (or importing) a working map of the planet that can tell us if the particular spot being observed is indeed that planet. Some other challenges would include the presence of asteroids or other celestial objects or the large number of rocks present near the planet that would cloud our observations.

The antenna would have to have a gain high enough to provide the necessary resolution to be able to observe any disturbances whatsoever which creates a challenge of its own so that is definitely something to look out for.

The full extent of how well the observations would turn out can be determined by carrying out observations of our own and comparing them with the official Jove kit observations by various individuals across the planet.

## Software used

- Ansys HFSS was to be used in the entire project for simulations of the antennas in mind.
- GNURadio, SDRConsoleV3

## Observations

- A 5 element Yagi-Uda antenna proved to be the only one that was usable for observation of such a range of frequencies.
- The dimensions of the Yagi-Uda turned out to be completely out of proportions as to what was to be constructed.
- An attempt to address the size issue of the antenna was addressed by installing interconnects in the design, which solved the issue of size but the gain was totally lost due to the losses at the interconnects.
- No further improvements sounded feasible at the point so the decision to order the Jove kit in its entirety was made to do the observations.

## Objectives reached

- Understanding the working of antennas and learning the different parameters and radiation patterns in order to select the ideal antenna for the project.
- Installing Ansys HFSS and learning how to design, simulate and analyse antennas
- Designing the 5 element Yagi-Uda antenna for the observations.

## Future Work

(Objectives yet to be achieved)

## References


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